EXERCISE 1

a We seek the tangent to $y = x - 2x^2 + 3$ at $x = 2$.

When $x = 2$, $y = 2 - 2(2)^2 + 3 = -3$

$\therefore$ the point of contact is $(2, -3)$.

Now $\dfrac{dy}{dx} = 1 - 4x$, so at $x = 2$,

$\dfrac{dy}{dx} = 1 - 8 = -7$

$\therefore$ the tangent has equation

$$\frac{y-(-3)}{x-2} = -7$$

$$\therefore\ y + 3 = -7(x-2)$$

$$\therefore\ y = -7x + 14 - 3$$

$$\therefore\ y = -7x + 11$$

b We seek the tangent to $y = \sqrt{x} + 1 = x^{\frac{1}{2}} + 1$ at $x = 4$.

When $x = 4$, $y = \sqrt{4} + 1 = 3$

$\therefore$ the point of contact is $(4, 3)$.

Now $\dfrac{dy}{dx} = \dfrac{1}{2\sqrt{x}}$, so at $x = 4$,

$\dfrac{dy}{dx} = \dfrac{1}{2\sqrt{4}} = \frac{1}{4}$

$\therefore$ the tangent has equation

$$\frac{y-3}{x-4} = \frac{1}{4}$$

$$\therefore\ 4y - 12 = x - 4$$

$$\therefore\ 4y = x + 8$$

c We seek the tangent to $y = x^3 - 5x$ at $x = 1$.

When $x = 1$, $y = 1^3 - 5(1) = -4$

$\therefore$ the point of contact is $(1, -4)$.

Now $\dfrac{dy}{dx} = 3x^2 - 5$, so at $x = 1$,

$\dfrac{dy}{dx} = 3 - 5 = -2$

$\therefore$ the tangent has equation

$$\frac{y-(-4)}{x-1} = -2$$

$$\therefore\ y + 4 = -2x + 2$$

$$\therefore\ y = -2x - 2$$

d We seek the tangent to $y = \dfrac{4}{\sqrt{x}}$ at $(1, 4)$.

Now $y = \dfrac{4}{\sqrt{x}} = 4x^{-\frac{1}{2}}$

$\therefore\ \dfrac{dy}{dx} = -2x^{-\frac{3}{2}}$ so at $x = 1$,

$\dfrac{dy}{dx} = -2\left(1^{-\frac{3}{2}}\right) = -2$

$\therefore$ the tangent has equation

$$\frac{y-4}{x-1} = -2$$

$$\therefore\ y - 4 = -2x + 2$$

$$\therefore\ y = -2x + 6$$

e We seek the tangent to $y = \dfrac{3}{x} - \dfrac{1}{x^2} = 3x^{-1} - x^{-2}$ at $(-1, -4)$.

Now $\dfrac{dy}{dx} = -3x^{-2} + 2x^{-3}$

$= -\dfrac{3}{x^2} + \dfrac{2}{x^3}$ so at $(-1, -4)$.

$\dfrac{dy}{dx} = -\dfrac{3}{(-1)^2} + \dfrac{2}{(-1)^3}$

$= -3 - 2$

$= -5$

$\therefore$ the tangent has equation

$$\frac{y-(-4)}{x-(-1)} = -5$$

$$\therefore\ y + 4 = -5x - 5$$

$$\therefore\ y = -5x - 9$$

f We seek the tangent to $y = 3x^2 - \dfrac{1}{x} = 3x^2 - x^{-1}$ at $x = -1$.

When $x = -1$, $y = 3(-1)^2 - \frac{1}{(-1)} = 4$

$\therefore$ the point of contact is $(-1, 4)$.

Now $\dfrac{dy}{dx} = 6x + x^{-2}$

$= 6x + \dfrac{1}{x^2}$ so at $x = -1$,

$\dfrac{dy}{dx} = 6(-1) + \dfrac{1}{(-1)^2} = -5$

$\therefore$ the tangent has equation

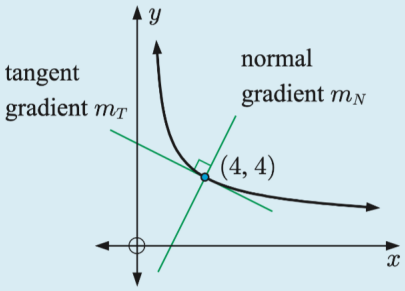
$$\frac{y-4}{x-(-1)} = -5$$

$$\therefore\ y - 4 = -5x - 5$$

$$\therefore\ y = -5x - 1$$

EXERCISE 2

When $x = 4$, $y = \dfrac{8}{\sqrt{4}} = \dfrac{8}{2} = 4$. So, the point of contact is $(4, 4)$.



Now as $y = 8x^{-\frac{1}{2}}$, $\dfrac{dy}{dx} = -4x^{-\frac{3}{2}}$

$\therefore$ when $x = 4$, $m_T = -4 \times 4^{-\frac{3}{2}} = -\frac{1}{2}$

$\therefore$ the normal at $(4, 4)$ has gradient $m_N = \frac{2}{1}$.

$\therefore$ the equation of the normal is

$$2x - 1y = 2(4) - 1(4)$$

or $2x - y = 4$

EXERCISE 3

a We seek the normal to $y = x^2$ at $(3, 9)$.

Now $\dfrac{dy}{dx} = 2x$ so at $x = 3$,

$\dfrac{dy}{dx} = 2(3) = 6 = \frac{6}{1}$

$\therefore$ the normal at $(3, 9)$ has gradient $-\frac{1}{6}$, so the equation of the normal is

$$\frac{y-9}{x-3} = -\frac{1}{6}$$

$\therefore \quad 6y - 54 = -x + 3$

$\therefore \quad 6y = -x + 57$

b We seek the normal to $y = x^3 - 5x + 2$ at $x = -2$.

When $x = -2$, $y = (-2)^3 - 5(-2) + 2 = 4$

and so the point of contact is $(-2, 4)$.

Now $\dfrac{dy}{dx} = 3x^2 - 5$ so at $x = -2$,

$\dfrac{dy}{dx} = 3(-2)^2 - 5 = 7$

$\therefore$ the normal at $(-2, 4)$ has gradient $-\frac{1}{7}$, so the equation of the normal is

$$\frac{y-4}{x-(-2)} = -\frac{1}{7}$$

$\therefore \quad 7y - 28 = -(x + 2)$

$\therefore \quad 7y = -x + 26$

c We seek the normal to $y = \dfrac{5}{\sqrt{x}} - \sqrt{x}$ at $(1, 4)$.

Now $y = 5x^{-\frac{1}{2}} - x^{\frac{1}{2}}$

$\therefore \dfrac{dy}{dx} = -\frac{5}{2}x^{-\frac{3}{2}} - \frac{1}{2}x^{-\frac{1}{2}}$ so at $x = 1$,

$\dfrac{dy}{dx} = -\frac{5}{2}\left(1^{-\frac{3}{2}}\right) - \frac{1}{2}\left(1^{-\frac{1}{2}}\right)$

$= -\frac{5}{2} - \frac{1}{2} = -3$

$\therefore$ the normal at $(1, 4)$ has gradient $\frac{1}{3}$, so the equation of the normal is

$$\frac{y-4}{x-1} = \frac{1}{3}$$

$\therefore \quad 3y - 12 = x - 1$

$\therefore \quad 3y = x + 11$

d We seek the normal to $y = 8\sqrt{x} - \dfrac{1}{x^2}$ at $x = 1$.

When $x = 1$, $y = 8\sqrt{1} - \dfrac{1}{1^2} = 7$

$\therefore$ the point of contact is $(1, 7)$.

Now $y = 8\sqrt{x} - \dfrac{1}{x^2} = 8x^{\frac{1}{2}} - x^{-2}$

$\therefore \dfrac{dy}{dx} = 4x^{-\frac{1}{2}} + 2x^{-3}$ so at $x = 1$,

$\dfrac{dy}{dx} = 4 + 2 = 6$

$\therefore$ the normal at $(1, 7)$ has gradient $-\frac{1}{6}$, so the equation of the normal is

$$\frac{y-7}{x-1} = -\frac{1}{6}$$

$\therefore \quad 6y - 42 = -x + 1$

$\therefore \quad 6y = -x + 43$

EXERCISE 4

Since $y = x^3 - 12x + 2$, $\dfrac{dy}{dx} = 3x^2 - 12$

Horizontal tangents have gradient 0,

so $3x^2 - 12 = 0$

$\therefore\ 3(x^2 - 4) = 0$

$\therefore\ 3(x+2)(x-2) = 0$

$\therefore\ x = -2$ or 2

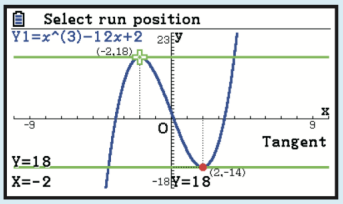
When $x = 2$, $y = 8 - 24 + 2 = -14$

When $x = -2$, $y = -8 + 24 + 2 = 18$

$\therefore$ the points of contact are $(2, -14)$ and $(-2, 18)$

$\therefore$ the tangents are $y = -14$ and $y = 18$.

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EXERCISE 5

$$y = 2x^3 + 3x^2 - 12x + 1$$

$$\therefore\ \frac{dy}{dx} = 6x^2 + 6x - 12$$

Horizontal tangents have gradient $= 0$

so $6x^2 + 6x - 12 = 0$

$\therefore\ x^2 + x - 2 = 0$

$\therefore\ (x+2)(x-1) = 0$

$\therefore\ x = -2$ or $x = 1$

Now at $x = -2$,

$y = 2(-2)^3 + 3(-2)^2 - 12(-2) + 1$

$= 21$

and at $x = 1$,

$y = 2(1)^3 + 3(1)^2 - 12(1) + 1$

$= -6$

$\therefore$ the points of contact are $(-2, 21)$ and $(1, -6)$

$\therefore$ the tangents are $y = 21$ and $y = -6$.

EXERCISE 6

Now $y = 2\sqrt{x} + \dfrac{1}{\sqrt{x}} = 2x^{\frac{1}{2}} + x^{-\frac{1}{2}}$

$\therefore\ \dfrac{dy}{dx} = x^{-\frac{1}{2}} - \frac{1}{2}x^{-\frac{3}{2}} = \dfrac{1}{\sqrt{x}} - \dfrac{1}{2x\sqrt{x}}$

Horizontal tangents have gradient $= 0$

$\therefore\ \dfrac{1}{\sqrt{x}} - \dfrac{1}{2x\sqrt{x}} = 0$

$\therefore\ \dfrac{2x-1}{2x\sqrt{x}} = 0$

$\therefore\ 2x - 1 = 0$

$\therefore\ x = \frac{1}{2}$

Now at $x = \frac{1}{2}$,

$$y = 2\sqrt{\tfrac{1}{2}} + \frac{1}{\sqrt{\frac{1}{2}}} = \frac{2\left(\frac{1}{2}\right) + 1}{\sqrt{\frac{1}{2}}} = \frac{2}{\sqrt{\frac{1}{2}}}$$

$$= 2\sqrt{2}$$

$\therefore$ the only horizontal tangent touches at the curve at $(\frac{1}{2}, 2\sqrt{2})$.

EXERCISE 7

Now $y = 2x^3 + kx^2 - 3$

$\therefore \quad \dfrac{dy}{dx} = 6x^2 + 2kx$

When $x = 2, \quad \dfrac{dy}{dx} = 4$

$\therefore \quad 6(2)^2 + 2k(2) = 4$

$\therefore \quad 24 + 4k = 4$

$\therefore \quad 4k = -20$

$\therefore \quad k = -5$

EXERCISE 8

Now $y = 1 - 3x + 12x^2 - 8x^3$

$\therefore \quad \dfrac{dy}{dx} = -3 + 24x - 24x^2$

When $x = 1, \quad \dfrac{dy}{dx} = -3 + 24 - 24 = -3$

$\therefore$ the tangent at $(1, 2)$ has gradient -3

The tangents to the curve have gradient -3 when $-3 + 24x - 24x^2 = -3$

$\therefore \quad 24x^2 - 24x = 0$

$\therefore \quad 24x(x - 1) = 0$

$\therefore$ when $x = 0$ or $x = 1$

So the other x-value for which the tangent to the curve has gradient -3 is $x = 0$,

and when $x = 0, \quad y = 1 - 0 + 0 - 0 = 1$

$\therefore$ the tangent to the curve at $(0, 1)$ is parallel to the tangent at $(1, 2)$.

This tangent has equation $\dfrac{y - 1}{x - 0} = -3$

or $y = -3x + 1$.

EXERCISE 9

a Now $y = x^2 + ax + b$

$\therefore \frac{dy}{dx} = 2x + a$

At $x = 1$, $\frac{dy}{dx} = 2 + a$

$\therefore$ the gradient of the tangent to the curve at $x = 1$ will be $2 + a$.

However the equation of the tangent is $2x + y = 6$ or $y = -2x + 6$

and so the gradient of the tangent is -2.

$\therefore 2 + a = -2$

$\therefore a = -4$

So, the curve is $y = x^2 - 4x + b$.

We also know that the tangent contacts the curve when $x = 1$.

$\therefore 1^2 - 4(1) + b = -2(1) + 6$

$\therefore 1 - 4 + b = 4$

$\therefore b = 7$

$\therefore a = -4, \quad b = 7$

c $y = 2x^2 - 1$

$\therefore \frac{dy}{dx} = 4x$

$\therefore$ at the point where $x = a$, $\frac{dy}{dx} = 4a$

$\therefore$ the gradient of the tangent at the point where $x = a$ is $4a$.

Also, at $x = a$, $y = 2a^2 - 1$.

$\therefore$ the tangent has equation

$$\frac{y - (2a^2 - 1)}{x - a} = 4a$$

$$\therefore y - 2a^2 + 1 = 4a(x - a)$$

$$\therefore y - 2a^2 + 1 = 4ax - 4a^2$$

$$\therefore 4ax - y = 2a^2 + 1$$

b Now $y = a\sqrt{x} + \frac{b}{\sqrt{x}} = ax^{\frac{1}{2}} + bx^{-\frac{1}{2}}$

$\therefore \frac{dy}{dx} = \frac{a}{2}x^{-\frac{1}{2}} - \frac{b}{2}x^{-\frac{3}{2}}$

$\therefore$ at $x = 4$, $\frac{dy}{dx} = \frac{a}{2}\left(4^{-\frac{1}{2}}\right) - \frac{b}{2}\left(4^{-\frac{3}{2}}\right)$

$$= \frac{a}{2}\left(\frac{1}{2}\right) - \frac{b}{2}\left(\frac{1}{8}\right)$$

$$= \frac{a}{4} - \frac{b}{16}$$

$\therefore$ the gradient of the tangent to the curve at $x = 4$ will be $\frac{a}{4} - \frac{b}{16} = \frac{4a - b}{16}$

However the equation of the *normal* is $4x + y = 22$ or $y = -4x + 22$.

$\therefore$ the normal has gradient -4.

$\therefore$ the tangent has gradient $\frac{1}{4}$, and so

$$\frac{4a - b}{16} = \frac{1}{4}$$

$$\therefore 4a - b = 4$$

$$\therefore b = 4a - 4 \quad \ldots. (1)$$

Also, at $x = 4$ the normal line intersects the curve.

$$\therefore a\sqrt{4} + \frac{b}{\sqrt{4}} = -4(4) + 22$$

$$\therefore 2a + \frac{b}{2} = 6$$

Consequently, $2a + \frac{4a - 4}{2} = 6$ {using (1)}

$$\therefore 2a + 2a - 2 = 6$$

$$\therefore 4a = 8$$

$$\therefore a = 2$$

and so $b = 4(2) - 4 = 4$ {from (1)}

EXERCISE 10

a $y = \sqrt{2x+1}$

When $x = 4$, $y = \sqrt{2(4)+1} = 3$,
so the point of contact is $(4, 3)$.

Now $\dfrac{dy}{dx} = \frac{1}{2}(2x+1)^{-\frac{1}{2}}(2) = \dfrac{1}{\sqrt{2x+1}}$

$\therefore$ at $x = 4$, $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{2(4)+1}} = \frac{1}{3}$

$\therefore$ the tangent has equation $\dfrac{y-3}{x-4} = \frac{1}{3}$
or $3y = x + 5$.

b $y = \dfrac{1}{2-x} = (2-x)^{-1}$

$\therefore$ at $x = -1$, $y = \dfrac{1}{2-(-1)} = \frac{1}{3}$

So the point of contact is $(-1, \frac{1}{3})$.

Now $\dfrac{dy}{dx} = -1(2-x)^{-2}(-1) = \dfrac{1}{(2-x)^2}$

$\therefore$ at $x = -1$, $\dfrac{dy}{dx} = \dfrac{1}{(2-(-1))^2} = \frac{1}{9}$

$\therefore$ the tangent has equation

$$\frac{y-\frac{1}{3}}{x-(-1)} = \frac{1}{9}$$

$$\therefore \quad 9y - 3 = x + 1$$

$$\therefore \quad 9y = x + 4$$

c We seek the tangent to $f(x) = \dfrac{x}{1-3x}$ at $(-1, -\frac{1}{4})$.

$f(x)$ is a quotient where

$u = x$ and $v = 1 - 3x$

$\therefore$ $u' = 1$ and $v' = -3$

Now $f'(x) = \dfrac{u'v - uv'}{v^2}$ {quotient rule}

$$\therefore \quad f'(x) = \frac{1(1-3x) - x(-3)}{(1-3x)^2}$$

$$= \frac{1}{(1-3x)^2}$$

$$\therefore \quad f'(-1) = \frac{1}{(1-3(-1))^2} = \frac{1}{16}$$

$\therefore$ the tangent has equation

$$\frac{y-\left(-\frac{1}{4}\right)}{x-(-1)} = \frac{1}{16}$$

$$\therefore \quad 16y + 4 = x + 1$$

$$\therefore \quad 16y = x - 3$$

d We seek the tangent to $f(x) = \dfrac{x^2}{1-x}$ at $(2, -4)$.

$f(x)$ is a quotient where

$u = x^2$ and $v = 1 - x$

$\therefore$ $u' = 2x$ and $v' = -1$

Now $f'(x) = \dfrac{u'v - uv'}{v^2}$ {quotient rule}

$$\therefore \quad f'(x) = \frac{2x(1-x) - x^2(-1)}{(1-x)^2}$$

$$= \frac{2x - 2x^2 + x^2}{(1-x)^2} = \frac{2x - x^2}{(1-x)^2}$$

$$\therefore \quad f'(2) = \frac{2(2) - 2^2}{(1-2)^2} = \frac{4-4}{1} = 0$$

As the tangent has gradient 0, it is horizontal.
$\therefore$ its equation is $y = c$
Since the contact point is $(2, -4)$, the tangent has equation $y = -4$.

EXERCISE 11

a We seek the normal to $y = \dfrac{1}{(x^2+1)^2}$ at $(1, \frac{1}{4})$.

As $y = (x^2+1)^{-2}$,

$$\frac{dy}{dx} = -2(x^2+1)^{-3}(2x) = \frac{-4x}{(x^2+1)^3}$$

$\therefore$ at $x = 1$, $\dfrac{dy}{dx} = \dfrac{-4}{(1+1)^3} = \dfrac{-4}{8} = -\frac{1}{2}$

$\therefore$ the normal at $(1, \frac{1}{4})$ has gradient 2.

So the equation of the normal is

$$\frac{y-\frac{1}{4}}{x-1} = 2$$

$$\therefore \quad y - \tfrac{1}{4} = 2x - 2$$

$$\therefore \quad y = 2x - \tfrac{7}{4}$$

b $y = \dfrac{1}{\sqrt{3-2x}}$

$\therefore$ at $x = -3$, $y = \dfrac{1}{\sqrt{3-2(-3)}} = \frac{1}{3}$

$\therefore$ the point of contact is $(-3, \frac{1}{3})$

Now $y = (3-2x)^{-\frac{1}{2}}$

$$\therefore \quad \frac{dy}{dx} = -\tfrac{1}{2}(3-2x)^{-\frac{3}{2}}(-2) = (3-2x)^{-\frac{3}{2}}$$

$\therefore$ at $x = -3$, $\dfrac{dy}{dx} = (3-2(-3))^{-\frac{3}{2}}$

$$= 9^{-\frac{3}{2}} = 3^{-3} = \tfrac{1}{27}$$

$\therefore$ the normal at $(-3, \frac{1}{3})$ has gradient -27.

So the equation of the normal is

$$\frac{y-\frac{1}{3}}{x-(-3)} = -27$$

$$\therefore \quad y - \tfrac{1}{3} = -27(x+3)$$

$$\therefore \quad y = -27x - \tfrac{242}{3}$$

c $f(x) = \sqrt{x}(1-x)^2$

Since $f(4) = \sqrt{4}(1-4)^2 = 18$,

the point of contact is $(4, 18)$.

Now $f(x)$ is a product where

$u = x^{\frac{1}{2}}$ and $v = (1-x)^2$

$\therefore$ $u' = \frac{1}{2}x^{-\frac{1}{2}}$ and $v' = 2(1-x)(-1)$

$= -2(1-x)$

Now $f'(x) = u'v + uv'$ {product rule}

$$\therefore \quad f'(x) = \tfrac{1}{2}x^{-\frac{1}{2}}(1-x)^2 - x^{\frac{1}{2}}2(1-x)$$

$$\therefore \quad f'(4) = \tfrac{1}{2\sqrt{4}}(1-4)^2 - \sqrt{4}(2)(1-4)$$

$$= \tfrac{1}{4}(9) - 2(2)(-3) = \tfrac{57}{4}$$

$\therefore$ the normal at $(4, 18)$ has gradient $-\frac{4}{57}$.

So, the equation of the normal is

$$\frac{y-18}{x-4} = -\tfrac{4}{57}$$

$$\therefore \quad 57(y-18) = -4(x-4)$$

$$\therefore \quad 57y = -4x + 1042$$

d $f(x) = \dfrac{x^2-1}{2x+3}$

Since $f(-1) = \dfrac{(-1)^2-1}{2(-1)+3} = \dfrac{0}{1} = 0$,

the point of contact is $(-1, 0)$.

Now $f(x)$ is a quotient where

$u = x^2 - 1$ and $v = 2x + 3$

$\therefore$ $u' = 2x$ and $v' = 2$

Now $f'(x) = \dfrac{u'v - uv'}{v^2}$

$$= \frac{2x(2x+3) - (x^2-1)(2)}{(2x+3)^2}$$

$\therefore$ $f'(-1)$

$$= \frac{2(-1)(-2+3) - ((-1)^2-1)(2)}{(2(-1)+3)^2}$$

$$= \frac{-2(1) - (0)(2)}{(1)^2} = -2$$

$\therefore$ the normal at $(-1, 0)$ has gradient $\frac{1}{2}$.

So, the equation of the normal is

$$\frac{y-0}{x-(-1)} = \tfrac{1}{2}$$

or $2y = x + 1$

EXERCISE 12

a $f(x) = e^{-x}$

$\therefore\ f(1) = e^{-1}$

$\therefore$ the point of contact is $\left(1, \frac{1}{e}\right)$.

Now $f'(x) = -e^{-x}$

$\therefore\ f'(1) = -e^{-1} = -\frac{1}{e}$

So, the gradient of the tangent is $-\frac{1}{e}$

$\therefore$ the tangent has equation $\frac{y - \frac{1}{e}}{x - 1} = -\frac{1}{e}$

$\therefore\ e\left(y - \frac{1}{e}\right) = -(x - 1)$

$\therefore\ ey - 1 = -x + 1$

$\therefore\ x + ey = 2$

or $y = -\frac{1}{e}x + \frac{2}{e}$

b $y = \ln(2 - x)$

so when $x = -1$, $y = \ln 3$

$\therefore$ the point of contact is $(-1,\ \ln 3)$.

Now $\frac{dy}{dx} = \frac{-1}{2 - x}$

$\therefore$ when $x = -1$, $\frac{dy}{dx} = -\frac{1}{2+1} = -\frac{1}{3}$

So, the gradient of the tangent is $-\frac{1}{3}$.

$\therefore$ the tangent has equation $\frac{y - \ln 3}{x + 1} = -\frac{1}{3}$

$\therefore\ 3(y - \ln 3) = -(x + 1)$

$\therefore\ 3y - 3\ln 3 = -x - 1$

$\therefore\ x + 3y = 3\ln 3 - 1$

c $y = \ln\sqrt{x}$

$= \ln x^{\frac{1}{2}}$

$= \frac{1}{2}\ln x$

$\therefore$ when $y = -1$, $-1 = \frac{1}{2}\ln x$

$\therefore\ \ln x = -2$

$\therefore\ x = e^{-2}$

$\therefore\ x = \frac{1}{e^2}$

$\therefore$ the point of contact is $\left(\frac{1}{e^2}, -1\right)$

Now $\frac{dy}{dx} = \frac{1}{2}\frac{1}{x} = \frac{1}{2x}$, so at the point of contact, $\frac{dy}{dx} = \frac{1}{2e^{-2}} = \frac{e^2}{2}$

$\therefore$ the tangent has gradient $\frac{e^2}{2}$ and the normal has gradient $-\frac{2}{e^2}$

$\therefore$ the normal has equation $\frac{y + 1}{x - \frac{1}{e^2}} = -\frac{2}{e^2}$

$\therefore\ e^2(y + 1) = -2\left(x - \frac{1}{e^2}\right)$

$\therefore\ e^2y + e^2 = -2x + \frac{2}{e^2}$

$\therefore\ 2x + e^2y = \frac{2}{e^2} - e^2$ or $y = -\frac{2}{e^2}x + \frac{2}{e^4} - 1$

EXERCISE 13

When $x = \frac{\pi}{4}$, $y = \tan\frac{\pi}{4} = 1$

$\therefore$ the point of contact is $(\frac{\pi}{4}, 1)$.

Now $f(x) = \tan x$ has derivative $f'(x) = \dfrac{1}{\cos^2 x}$

$\therefore$ the tangent at $(\frac{\pi}{4}, 1)$ has gradient $\dfrac{1}{\cos^2\frac{\pi}{4}} = \dfrac{1}{\left(\frac{1}{\sqrt{2}}\right)^2} = 2$

$\therefore$ the tangent has equation $\dfrac{y-1}{x-\frac{\pi}{4}} = 2$

which is $y - 1 = 2x - \frac{\pi}{2}$

or $y = 2x + (1 - \frac{\pi}{2})$

EXERCISE 14

a $y = \sin x \quad \therefore \quad \dfrac{dy}{dx} = \cos x$

When $x = 0$, $\dfrac{dy}{dx} = \cos 0 = 1$

$\therefore$ the tangent has equation $\dfrac{y-0}{x-0} = 1$

or $y = x$

b $y = \tan x \quad \therefore \quad \dfrac{dy}{dx} = \dfrac{1}{\cos^2 x}$

When $x = 0$, $\dfrac{dy}{dx} = \dfrac{1}{\cos^2 0} = 1$

$\therefore$ the tangent has equation $\dfrac{y-0}{x-0} = 1$

or $y = x$

c $y = \cos x \quad \therefore \quad \dfrac{dy}{dx} = -\sin x$

When $x = \frac{\pi}{6}$, $y = \frac{\sqrt{3}}{2}$

and $\dfrac{dy}{dx} = -\sin\left(\frac{\pi}{6}\right) = -\frac{1}{2}$

So, the normal has gradient 2,

and its equation is $\dfrac{y-\frac{\sqrt{3}}{2}}{x-\frac{\pi}{6}} = 2$

$\therefore \quad y - \frac{\sqrt{3}}{2} = 2x - \frac{\pi}{3}$

$\therefore \quad 2x - y = \frac{\pi}{3} - \frac{\sqrt{3}}{2}$

d $y = \dfrac{1}{\sin(2x)} = (\sin(2x))^{-1}$

$\therefore \quad \dfrac{dy}{dx} = -1(\sin(2x))^{-2} \times 2\cos(2x)$

$= -\dfrac{2\cos(2x)}{(\sin(2x))^2}$

When $x = \frac{\pi}{4}$, $y = 1$

and $\dfrac{dy}{dx} = -\dfrac{2\cos(\frac{\pi}{2})}{\left(\sin(\frac{\pi}{2})\right)^2} = 0$

$\therefore$ the gradient of the normal is undefined, so the normal is $x = \frac{\pi}{4}$.

EXERCISE 15

a Consider the tangent to $y = x^3$ at $x = 2$.
When $x = 2$, $y = 2^3 = 8$ so the point of contact is $(2, 8)$

Now $\dfrac{dy}{dx} = 3x^2$ and so at $x = 2$,

$\dfrac{dy}{dx} = 3(2)^2 = 12$

$\therefore$ the tangent at $(2, 8)$ has gradient 12 and its equation is $\dfrac{y-8}{x-2} = 12$

$\therefore \ y - 8 = 12x - 24$

$\therefore \ y = 12x - 16$

$\therefore$ the tangent meets the curve where

$12x - 16 = x^3$

$\therefore \ x^3 - 12x + 16 = 0$

Because the tangent touches the curve at $x = 2$, there must be a repeated solution at this point.

$\therefore \ (x-2)^2$ must be a factor of this cubic

$\therefore \ (x-2)^2(x+4) = 0$

$\therefore$ the tangent meets the curve again when $x = -4$.

When $x = -4$, $y = (-4)^3 = -64$

$\therefore$ the tangent meets the curve again at $(-4, -64)$.

b Consider the tangent to $y = -x^3 + 2x^2 + 1$ at $x = -1$.
When $x = -1$, $y = -(-1)^3 + 2(-1)^2 + 1$
$= 4$
and so the point of contact is $(-1, 4)$.

Now $\dfrac{dy}{dx} = -3x^2 + 4x$ and so at $x = -1$,

$\dfrac{dy}{dx} = -3(-1)^2 + 4(-1) = -7$

$\therefore$ the tangent at $(-1, 4)$ has gradient -7 and its equation is $\dfrac{y-4}{x-(-1)} = -7$

$\therefore \ y - 4 = -7(x+1)$

$\therefore \ y = -7x - 3$

$\therefore$ the tangent meets the curve where

$-7x - 3 = -x^3 + 2x^2 + 1$

$\therefore \ x^3 - 2x^2 - 7x - 4 = 0$

Because the tangent touches the curve at $x = -1$, there must be a repeated solution at this point.

$\therefore \ (x+1)^2$ must be a factor of this cubic

$\therefore \ (x+1)^2(x-4) = 0$

$\therefore$ the tangent meets the curve again when $x = 4$.

When $x = 4$, $y = -(4)^3 + 2(4)^2 + 1$
$= -64 + 32 + 1 = -31$

$\therefore$ the tangent meets the curve again at $(4, -31)$.

EXERCISE 16

a $f(x) = x^2 + \dfrac{4}{x^2}$

$\therefore \ f'(x) = 2x - 2 \times \dfrac{4}{x^3}$

$\therefore \ f'(x) = 2x - \dfrac{8}{x^3}$

b Horizontal tangents have gradient 0, so

$2x - \dfrac{8}{x^3} = 0$

$\therefore \ 2x^4 = 8$

$\therefore \ x^4 = 4$

$\therefore \ x = \pm\sqrt{2}$

c When $x = -\sqrt{2}$,

$f(-\sqrt{2}) = \left(-\sqrt{2}\right)^2 + \dfrac{4}{\left(-\sqrt{2}\right)^2} = 2 + \dfrac{4}{2} = 4$

$\therefore$ the horizontal tangent at $(-\sqrt{2}, 4)$ is $y = 4$.

When $x = \sqrt{2}$,

$f(\sqrt{2}) = \left(\sqrt{2}\right)^2 + \dfrac{4}{\left(\sqrt{2}\right)^2} = 2 + \dfrac{4}{2} = 4$

$\therefore$ the horizontal tangent at $(\sqrt{2}, 4)$ is $y = 4$.

$\therefore$ the tangents are the same line because they have the same equation.

EXERCISE 17

$y = x^2e^x$ so when $x = 1,\ y = e$

$\therefore$ the point of contact is $(1, e)$.

Now $\dfrac{dy}{dx} = 2xe^x + x^2e^x$

$\therefore$ when $x = 1,\ \dfrac{dy}{dx} = 2e + e = 3e$

$\therefore$ the tangent has equation $\dfrac{y - e}{x - 1} = 3e$

$\therefore\ y - e = 3ex - 3e$

$\therefore\ y - 3ex = -2e$

$\therefore\ 3ex - y = 2e$

The tangent cuts the x-axis when

$y = 0$

$\therefore\ 3ex = 2e$

$\therefore\ x = \frac{2}{3}$

and the y-axis when

$x = 0$

$\therefore\ -y = 2e$

$\therefore\ y = -2e$

So, A is $(\frac{2}{3}, 0)$ and B is $(0, -2e)$.

EXERCISE 18

a Consider the tangent to $y = x^2 - x + 9$ at $x = a$.

When $x = a$, $y = a^2 - a + 9$, so the point of contact is $(a, a^2 - a + 9)$.

Now $\dfrac{dy}{dx} = 2x - 1$ and so at $x = a$, $\dfrac{dy}{dx} = 2a - 1$

$\therefore$ the gradient of the tangent at $(a, a^2 - a + 9)$ is $2a - 1$

$\therefore$ the equation of the tangent is $\dfrac{y - (a^2 - a + 9)}{x - a} = 2a - 1$

$$\therefore \quad y - (a^2 - a + 9) = (2a - 1)(x - a)$$
$$\therefore \quad y = (2a - 1)x - 2a^2 + a + a^2 - a + 9$$
$$\therefore \quad y = (2a - 1)x - a^2 + 9 \quad \ldots. (1)$$

But this tangent passes through $(0, 0)$, so $0 = a^2 - 9$

$$\therefore \quad (a + 3)(a - 3) = 0$$
$$\therefore \quad a = \pm 3$$

$\therefore$ the tangents are: At $a = 3$, $y = (2(3) - 1)x - 3^2 + 9$ {from (1)}

$$\therefore \quad y = 5x, \quad \text{with contact at } (3, 15).$$

At $a = -3$, $y = (2(-3) - 1)x - (-3)^2 + 9$ {from (1)}

$$\therefore \quad y = -7x, \quad \text{with contact at } (-3, 21).$$

b Let (a, a^3) lie on $y = x^3$.

Now $\dfrac{dy}{dx} = 3x^2$, so at $x = a$, $\dfrac{dy}{dx} = 3a^2$

$\therefore$ the gradient of the tangent at (a, a^3) is $3a^2$

$\therefore$ the equation of the tangent is $\dfrac{y - a^3}{x - a} = 3a^2$ or $y - a^3 = (3a^2)(x - a)$

But this tangent passes through $(-2, 0)$, so $0 - a^3 = 3a^2(-2 - a)$

$$\therefore \quad -a^3 = -6a^2 - 3a^3$$
$$\therefore \quad 2a^3 + 6a^2 = 0$$
$$\therefore \quad 2a^2(a + 3) = 0$$
$$\therefore \quad a = 0 \text{ or } -3$$

If $a = 0$, the tangent equation is $y = 0$, with contact point $(0, 0)$.

If $a = -3$, the tangent equation is $y - (-27) = 27(x + 3)$

$$\therefore \quad y = 27x + 54, \quad \text{with contact point } (-3, -27).$$

c Let $(a, \sqrt{a})$ lie on $y = \sqrt{x}$.

Now $\dfrac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}} = \dfrac{1}{2\sqrt{x}}$, so at $x = a$, $\dfrac{dy}{dx} = \dfrac{1}{2\sqrt{a}}$

$\therefore$ the gradient of the tangent at $(a, \sqrt{a})$ is $\dfrac{1}{2\sqrt{a}}$

and the gradient of the normal at this point is $-2\sqrt{a}$.

$\therefore$ the normal has equation $\dfrac{y-\sqrt{a}}{x-a} = -2\sqrt{a}$

or $y - \sqrt{a} = -2\sqrt{a}(x-a)$.

But this normal passes through $(4, 0)$, so $0 - \sqrt{a} = -2\sqrt{a}(4-a)$

$\therefore \quad 2\sqrt{a}(4-a) - \sqrt{a} = 0$

$\therefore \quad \sqrt{a}(8 - 2a - 1) = 0$

$\therefore \quad \sqrt{a}(7 - 2a) = 0$

$\therefore \quad a = 0$ or $\frac{7}{2}$

But $a = 0$ is the endpoint of the function, so there is no normal here.

When $a = \frac{7}{2}$, $\quad y - \sqrt{\frac{7}{2}} = -2\sqrt{\frac{7}{2}}\left(x - \frac{7}{2}\right)$

$\therefore \quad \sqrt{2}y - \sqrt{7} = -2\sqrt{7}\left(x - \frac{7}{2}\right)$

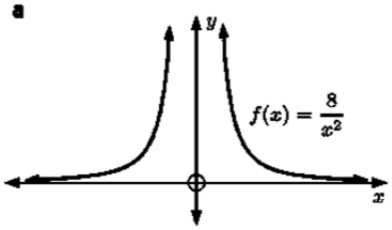
$\therefore \quad \sqrt{2}y + 2\sqrt{7}x = 7\sqrt{7} + \sqrt{7}$

$\therefore \quad \sqrt{2}y + 2\sqrt{7}x = 8\sqrt{7}$

$\therefore \quad y = -\sqrt{14}x + 4\sqrt{14}$ with contact point $(\frac{7}{2}, \sqrt{\frac{7}{2}})$.

EXERCISE 19

a



b Let $\left(a, \dfrac{8}{a^2}\right)$ lie on $f(x) = \dfrac{8}{x^2} = 8x^{-2}$

Now $f'(x) = -16x^{-3} = -\dfrac{16}{x^3}$

$\therefore \quad f'(a) = -\dfrac{16}{a^3}$

$\therefore$ the gradient of the tangent at $\left(a, \dfrac{8}{a^2}\right)$ is $-\dfrac{16}{a^3}$

$\therefore$ the equation of the tangent is $\dfrac{y - \frac{8}{a^2}}{x - a} = -\dfrac{16}{a^3}$

$\therefore \quad a^3y - 8a = -16x + 16a$

$\therefore \quad 16x + a^3y = 24a$

c The tangent cuts the x-axis when $y = 0$

$\therefore \quad 16x = 24a$

$\therefore \quad x = \frac{3}{2}a$

$\therefore$ A is $(\frac{3}{2}a, 0)$.

The tangent cuts the y-axis when $x = 0$

$\therefore \quad a^3y = 24a$

$\therefore \quad y = \dfrac{24}{a^2}$

$\therefore$ B is $\left(0, \dfrac{24}{a^2}\right)$.

d Area of triangle OAB

$= \left|\frac{1}{2} \times \left(\frac{3}{2}a\right) \times \left(\dfrac{24}{a^2}\right)\right|$

$= \dfrac{18}{|a|}$ units^2

As $a \to \infty$, $\dfrac{18}{a} \to 0$

$\therefore$ area $\to 0$

EXERCISE 20

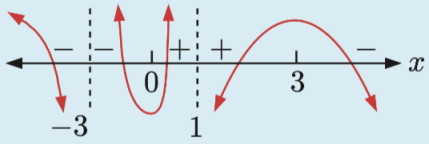
a $f(x) = \dfrac{2x-3}{x^2+2x-3}$

$\therefore\ f'(x) = \dfrac{2(x^2+2x-3)-(2x-3)(2x+2)}{(x^2+2x-3)^2}$ {quotient rule}

$= \dfrac{2x^2+4x-6-[4x^2-2x-6]}{((x-1)(x+3))^2}$

$= \dfrac{-2x^2+6x}{(x-1)^2(x+3)^2}$

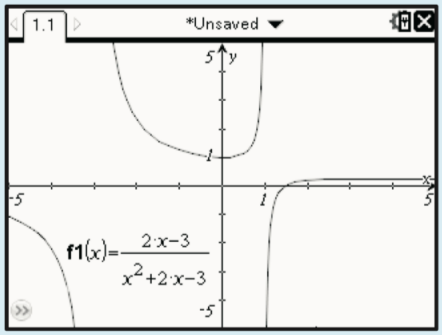
$= \dfrac{-2x(x-3)}{(x-1)^2(x+3)^2}$ which has sign diagram:



b $f(x)$ is increasing for $0 \leqslant x < 1$
and for $1 < x \leqslant 3$.

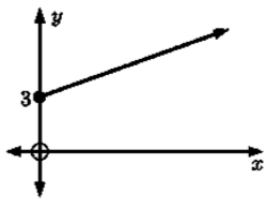
$f(x)$ is decreasing for $x < -3$
and for $-3 < x \leqslant 0$
and for $x \geqslant 3$.

TI-*n*spire



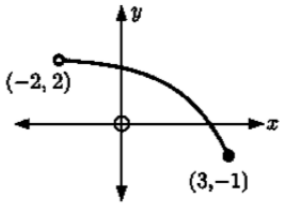
EXERCISE 21

a



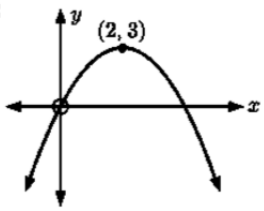
i $x \geqslant 0$ **ii** never

b



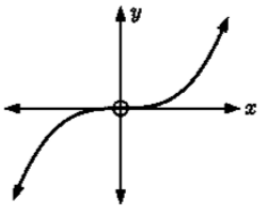
i never **ii** $-2 < x \leqslant 3$

c



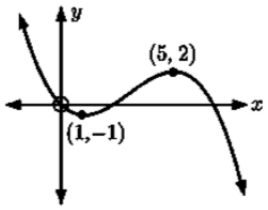
i $x \leqslant 2$ **ii** $x \geqslant 2$

d



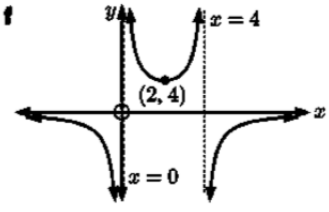
i all real x **ii** never

e



i $1 \leqslant x \leqslant 5$
ii $x \leqslant 1,\ x \geqslant 5$

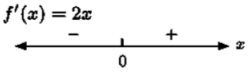
f



i $2 \leqslant x < 4,\ x > 4$
ii $x < 0,\ 0 < x \leqslant 2$

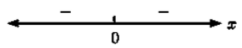
EXERCISE 22

a $f(x) = x^2, \quad f'(x) = 2x$

Sign diagram of $f'(x)$: 

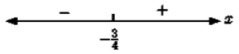
increasing when $x \geqslant 0$,
decreasing when $x \leqslant 0$

b $f(x) = -x^3, \quad f'(x) = -3x^2$

Sign diagram of $f'(x)$: 

decreasing for all x

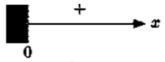
c $f(x) = 2x^2 + 3x - 4, \quad f'(x) = 4x + 3$

Sign diagram of $f'(x)$: 

increasing when $x \geqslant -\frac{3}{4}$,
decreasing when $x \leqslant -\frac{3}{4}$

d $f(x) = \sqrt{x} = x^{\frac{1}{2}}$,

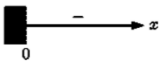
$f'(x) = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$

Sign diagram of $f'(x)$: 

$f(x)$ is only defined when $x \geqslant 0$
increasing when $x \geqslant 0$, never decreasing

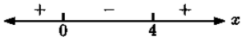
e $f(x) = \frac{2}{\sqrt{x}} = 2x^{-\frac{1}{2}}$

$f'(x) = -x^{-\frac{3}{2}} = \frac{-1}{x\sqrt{x}}$

Sign diagram of $f'(x)$: 

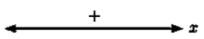
$f(x)$ is only defined for $x > 0$
never increasing, decreasing when $x > 0$

f $f(x) = x^3 - 6x^2, \quad f'(x) = 3x^2 - 12x$
$= 3x(x - 4)$

Sign diagram of $f'(x)$: 

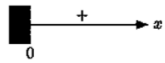
increasing when $x \leqslant 0$ or $x \geqslant 4$,
decreasing when $0 \leqslant x \leqslant 4$

g $f(x) = e^x, \quad f'(x) = e^x$

Sign diagram of $f'(x)$: 

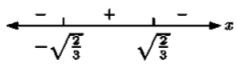
$f(x)$ is increasing for all x

h $f(x) = \ln x, \quad f'(x) = \frac{1}{x}$

Sign diagram of $f'(x)$: 

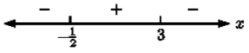
$f(x)$ is only defined when $x > 0$
increasing when $x > 0$, never decreasing

i $f(x) = -2x^3 + 4x$
$f'(x) = -6x^2 + 4$
$= -2(3x^2 - 2)$

Sign diagram of $f'(x)$: 

increasing for $-\sqrt{\frac{2}{3}} \leqslant x \leqslant \sqrt{\frac{2}{3}}$,
decreasing for $x \leqslant -\sqrt{\frac{2}{3}}$ or $x \geqslant \sqrt{\frac{2}{3}}$

j $f(x) = -4x^3 + 15x^2 + 18x + 3$
$f'(x) = -12x^2 + 30x + 18$
$= -6(2x^2 - 5x - 3)$
$= -6(2x + 1)(x - 3)$

Sign diagram of $f'(x)$: 

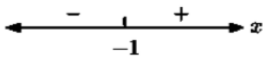
increasing when $-\frac{1}{2} \leqslant x \leqslant 3$,
decreasing when $x \leqslant -\frac{1}{2}$ or $x \geqslant 3$

k $f(x) = 3 + e^{-x}, \quad f'(x) = -e^{-x}$

Sign diagram of $f'(x)$: 

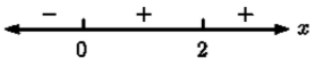
$f(x)$ is decreasing for all x.

l $f(x) = xe^x, \quad f'(x) = e^x + xe^x$
$= e^x(1 + x)$

Sign diagram of $f'(x)$: 

increasing when $x \geqslant -1$
decreasing when $x \leqslant -1$.

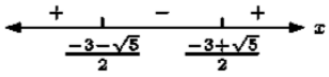
m $f(x) = 3x^4 - 16x^3 + 24x^2 - 2,$
$f'(x) = 12x^3 - 48x^2 + 48x$
$= 12x(x^2 - 4x + 4)$
$= 12x(x - 2)^2$

Sign diagram of $f'(x)$: 

increasing when $x \geqslant 0$,
decreasing when $x \leqslant 0$

n $f(x) = 2x^3 + 9x^2 + 6x - 7,$
$f'(x) = 6x^2 + 18x + 6$
$= 6(x^2 + 3x + 1)$

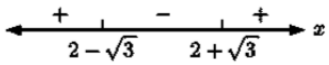
$f'(x) = 0$ when $x = \frac{-3 \pm \sqrt{9-4}}{2} = \frac{-3 \pm \sqrt{5}}{2}$

Sign diagram of $f'(x)$: 

increasing for $x \leqslant \frac{-3-\sqrt{5}}{2}$ or $x \geqslant \frac{-3+\sqrt{5}}{2}$,
decreasing for $\frac{-3-\sqrt{5}}{2} \leqslant x \leqslant \frac{-3+\sqrt{5}}{2}$

o $f(x) = x^3 - 6x^2 + 3x - 1,$
$f'(x) = 3x^2 - 12x + 3$
$= 3(x^2 - 4x + 1)$

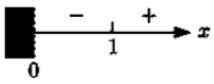
$f'(x) = 0$ when $x = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$

Sign diagram of $f'(x)$: 

increasing when $x \leqslant 2 - \sqrt{3}$
or $x \geqslant 2 + \sqrt{3}$,
decreasing when $2 - \sqrt{3} \leqslant x \leqslant 2 + \sqrt{3}$

p $f(x) = x - 2\sqrt{x} = x - 2x^{\frac{1}{2}}$

$f'(x) = 1 - x^{-\frac{1}{2}} = 1 - \frac{1}{\sqrt{x}} = \frac{\sqrt{x} - 1}{\sqrt{x}}$

Sign diagram of $f'(x)$: 

increasing when $x \geqslant 1$,
decreasing when $0 \leqslant x \leqslant 1$

EXERCISE 23

a $f(x) = \frac{4x}{x^2 + 1}$ is a quotient with

$u = 4x$ and $v = x^2 + 1$
$\therefore \quad u' = 4$ and $v' = 2x$

$\therefore \quad f'(x) = \frac{4(x^2 + 1) - 4x \times 2x}{(x^2 + 1)^2}$

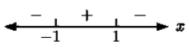
$= \frac{4x^2 + 4 - 8x^2}{(x^2 + 1)^2}$

$= \frac{4 - 4x^2}{(x^2 + 1)^2}$

$= \frac{-4(x^2 - 1)}{(x^2 + 1)^2}$

$= \frac{-4(x + 1)(x - 1)}{(x^2 + 1)^2}$

Sign diagram of $f'(x)$:



b $f(x)$ is increasing for $-1 \leqslant x \leqslant 1$,
decreasing for $x \leqslant -1$ and $x \geqslant 1$

EXERCISE 24

a $f(x) = \dfrac{4x}{(x-1)^2}$ is a quotient with

$u = 4x$ and $v = (x-1)^2$

$\therefore \quad u' = 4$ and $v' = 2(x-1)$

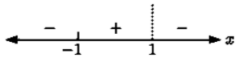
$$\therefore \quad f'(x) = \frac{4(x-1)^2 - 8x(x-1)}{(x-1)^4}$$

$$= \frac{4(x-1)[(x-1) - 2x]}{(x-1)^4}$$

$$= \frac{4(-1-x)}{(x-1)^3}$$

$$= \frac{-4(x+1)}{(x-1)^3}$$

Sign diagram of $f'(x)$:



b $f(x)$ is increasing for $-1 \leqslant x < 1$,
decreasing for $x \leqslant -1$ and $x > 1$

EXERCISE 25

a $f(x) = \dfrac{-x^2 + 4x - 7}{x - 1}$ is a quotient with $u = -x^2 + 4x - 7$ and $v = x - 1$

$\therefore \quad u' = -2x + 4$ and $v' = 1$

$$\therefore \quad f'(x) = \frac{(-2x+4)(x-1) - (-x^2+4x-7)(1)}{(x-1)^2}$$

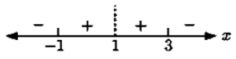
$$= \frac{-2x^2 + 6x - 4 + x^2 - 4x + 7}{(x-1)^2}$$

$$= \frac{-x^2 + 2x + 3}{(x-1)^2}$$

$$= \frac{-(x^2 - 2x - 3)}{(x-1)^2}$$

$$= \frac{-(x+1)(x-3)}{(x-1)^2}$$

Sign diagram of $f'(x)$:

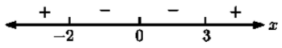


b $f(x)$ is increasing for $-1 \leqslant x < 1$
and $1 < x \leqslant 3$, and decreasing
for $x \leqslant -1$ and $x \geqslant 3$.

EXERCISE 26

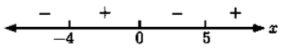
a A is a *local maximum*, B is a *stationary inflection*, C is a *local minimum*.

b $f'(x)$ has sign diagram:



c I $f(x)$ is increasing for $x \leqslant -2$ and $x \geqslant 3$ II $f(x)$ is decreasing for $-2 \leqslant x \leqslant 3$

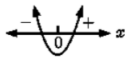
d $f(x)$ has sign diagram:



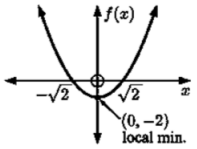
EXERCISE 27

a $f(x) = x^2 - 2 \quad \therefore \quad f'(x) = 2x$

with sign diagram:

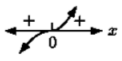


Now $f(0) = -2$,
so there is a local minimum at $(0, -2)$.

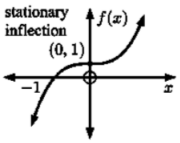


b $f(x) = x^3 + 1 \quad \therefore \quad f'(x) = 3x^2$

with sign diagram:



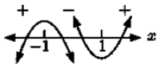
Now $f(0) = 1$,
so there is a stationary inflection at $(0, 1)$.



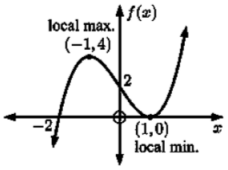
c $f(x) = x^3 - 3x + 2$

$\therefore \quad f'(x) = 3x^2 - 3$
$= 3(x^2 - 1)$
$= 3(x + 1)(x - 1)$

with sign diagram:



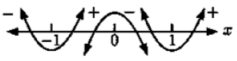
Now $f(-1) = 4$, $f(1) = 0$,
so there is a local maximum at $(-1, 4)$,
and a local minimum at $(1, 0)$.



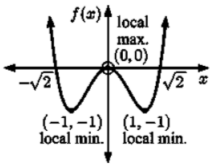
d $f(x) = x^4 - 2x^2$

$\therefore \quad f'(x) = 4x^3 - 4x$
$= 4x(x^2 - 1)$
$= 4x(x + 1)(x - 1)$

with sign diagram:



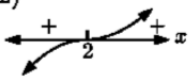
Now $f(-1) = -1$, $f(1) = -1$, $f(0) = 0$,
so there are local minima at $(-1, -1)$ and
$(1, -1)$, and a local maximum at $(0, 0)$.



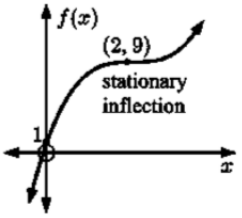
e $f(x) = x^3 - 6x^2 + 12x + 1$

$\therefore \quad f'(x) = 3x^2 - 12x + 12$
$= 3(x^2 - 4x + 4)$
$= 3(x - 2)^2$

with sign diagram:



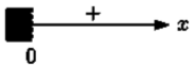
Now $f(2) = 9$, so there is a
stationary inflection at $(2, 9)$.



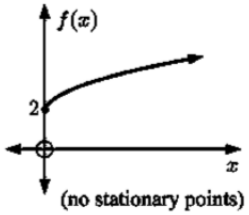
f $f(x) = \sqrt{x} + 2$

$\therefore \quad f'(x) = \frac{1}{2}x^{-\frac{1}{2}}$
$= \dfrac{1}{2\sqrt{x}} \neq 0$

with sign diagram:



$\therefore$ no stationary points.

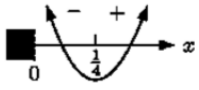


$f(x) = x - \sqrt{x}$

$\therefore \quad f'(x) = 1 - \frac{1}{2}x^{-\frac{1}{2}}$

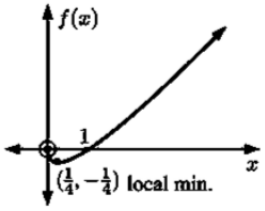
$= 1 - \dfrac{1}{2\sqrt{x}} = \dfrac{2\sqrt{x} - 1}{2\sqrt{x}}$

with sign diagram:



$f(x)$ is defined for all $x \geqslant 0$

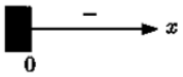
Now $f(\frac{1}{4}) = -\frac{1}{4}$, so there is a local minimum at $(\frac{1}{4}, -\frac{1}{4})$.



$f(x) = 1 - x\sqrt{x} = 1 - x^{\frac{3}{2}}$

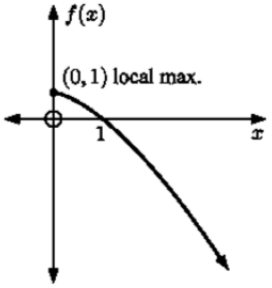
$\therefore \quad f'(x) = -\frac{3}{2}x^{\frac{1}{2}} = \dfrac{-3\sqrt{x}}{2}$

with sign diagram:



$f(x)$ is only defined when $x \geqslant 0$

Now $f(0) = 1$, so there is a local maximum at $(0, 1)$.



h $f(x) = x^4 - 6x^2 + 8x - 3$

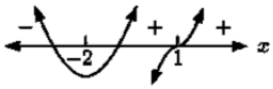
$\therefore \quad f'(x) = 4x^3 - 12x + 8$

$= 4(x^3 - 3x + 2)$

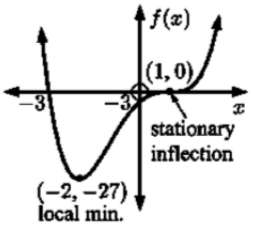
$= 4(x - 1)(x^2 + x - 2)$

$= 4(x - 1)(x + 2)(x - 1)$

with sign diagram:



Now $f(-2) = -27$, $f(1) = 0$, so there is a local minimum at $(-2, -27)$, and a stationary inflection at $(1, 0)$.

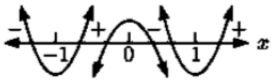


i $f(x) = x^4 - 2x^2 - 8$

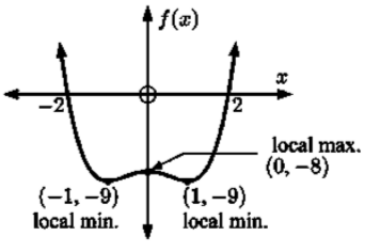
$\therefore \quad f'(x) = 4x^3 - 4x$

$= 4x(x^2 - 1) = 4x(x + 1)(x - 1)$

with sign diagram:



Now $f(-1) = -9$, $f(1) = -9$, $f(0) = -8$, so there are local minima at $(-1, -9)$ and $(1, -9)$, and a local maximum at $(0, -8)$.



EXERCISE 28

$f(x) = ax^2 + bx + c, \quad a \neq 0$

$\therefore \quad f'(x) = 2ax + b$

$f(x)$ has a stationary point when $f'(x) = 0$

$\therefore \quad x = -\dfrac{b}{2a}$

There is a local maximum when $a < 0$  and there is a local minimum when $a > 0$ 

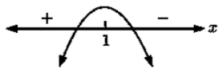
EXERCISE 29

a $y = xe^{-x}$

$\therefore \frac{dy}{dx} = 1e^{-x} - xe^{-x}$ {product rule}

$= e^{-x}(1-x)$

$= \frac{1-x}{e^x}$ which has sign diagram:

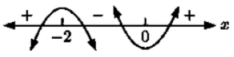


When $x = 1$, $y = 1e^{-1} = \frac{1}{e}$, so we have a local maximum at $\left(1, \frac{1}{e}\right)$.

b $y = x^2e^x$

$\therefore \frac{dy}{dx} = 2xe^x + x^2e^x$ {product rule}

$= xe^x(2+x)$ which has sign diagram:



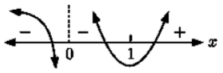
When $x = -2$, $y = 4e^{-2}$, and when $x = 0$, $y = 0$.

So, we have a local maximum at $\left(-2, \frac{4}{e^2}\right)$, and a local minimum at $(0, 0)$.

c $y = \frac{e^x}{x}$

$\therefore \frac{dy}{dx} = \frac{e^x x - e^x(1)}{x^2}$ {quotient rule}

$= \frac{e^x(x-1)}{x^2}$ which has sign diagram:



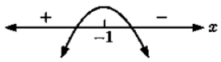
When $x = 1$, $y = \frac{e^1}{1} = e$, so we have a local minimum at $(1, e)$.

d $y = e^{-x}(x+2)$

$\therefore \frac{dy}{dx} = -e^{-x}(x+2) + e^{-x}$ {product rule}

$= e^{-x}(-x-2+1)$

$= e^{-x}(-x-1)$ which has sign diagram:



When $x = -1$, $y = e(-1+2) = e$, so we have a local maximum at $(-1, e)$.

EXERCISE 30

$f(x) = 2x^3 + ax^2 - 24x + 1$

$\therefore f'(x) = 6x^2 + 2ax - 24$

But $f'(-4) = 0$, so $96 - 8a - 24 = 0$

$\therefore 72 = 8a$

$\therefore a = 9$

EXERCISE 31

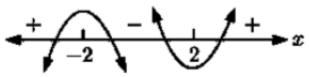


b Now $f(x) = x^3 - 12x - 13$

$\therefore f'(x) = 3x^2 - 12$

$= 3(x^2 - 4)$

$= 3(x+2)(x-2)$ with sign diagram:



Now $f(2) = -29$, so there is a local minimum at $(2, -29)$ and a local maximum at $(-2, 3)$.

EXERCISE 32

a $f(x)$ is defined when $\ln x$ is defined $\therefore$ $f(x)$ is defined for $x > 0$

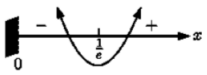
b $f'(x) = \ln x + \frac{x}{x}$ {product rule}

$= \ln x + 1$

which is 0 when $\ln x = -1$

$\therefore \quad x = e^{-1}$

Sign diagram of $f'(x)$ is:



So, there is a local minimum at $\left(\frac{1}{e}, \frac{1}{e}\ln\frac{1}{e}\right)$

$\therefore$ the global minimum value of $f(x)$ is $\frac{1}{e}\ln e^{-1} = -\frac{1}{e}$

EXERCISE 33

Consider $f(x) = \dfrac{\ln x}{x}$

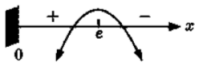
$\therefore \quad f'(x) = \dfrac{\left(\frac{1}{x}\right)x - \ln x(1)}{x^2} = \dfrac{1 - \ln x}{x^2}$

$\therefore \quad f'(x) = 0$ when $1 - \ln x = 0$

$\therefore \quad \ln x = 1$

$\therefore \quad x = e$

Sign diagram of $f'(x)$ is:



Now $f(e) = \dfrac{\ln e}{e} = \dfrac{1}{e}$

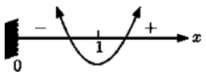
$\therefore$ there is a local maximum at $\left(e, \frac{1}{e}\right)$

$\therefore \quad f(x) \leqslant \frac{1}{e}$ for all x, and so $\dfrac{\ln x}{x} \leqslant \dfrac{1}{e}$ for all $x > 0$

EXERCISE 34

a $f(x) = x - \ln x$

$\therefore \quad f'(x) = 1 - \dfrac{1}{x} = \dfrac{x-1}{x}$ and the sign diagram of $f'(x)$ is:



$\therefore$ $f(x)$ has a local minimum at $(1, 1 - \ln 1)$ or $(1, 1)$. This is the only turning point.

b $f(x) \geqslant 1$ for all $x > 0$

$\therefore \quad x - \ln x \geqslant 1$

$\therefore \quad \ln x \leqslant x - 1$ for all $x > 0$

EXERCISE 35

a

A	+	−	+
B	−	0	+
C	+	+	0
D	+	0	−
E	0	−	−

b The turning points of $y = f(x)$ are point B, a local minimum, and point D, a local maximum.

c The inflection point of $y = f(x)$ is point C, a non-stationary point of inflection.

EXERCISE 36

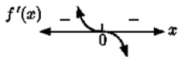
a $f(x) = x^2 + 3$

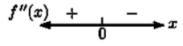
$\therefore\ f'(x) = 2x$

$\therefore\ f''(x) = 2$

Since $f''(x) \neq 0$,
no points of inflection exist.

b $f(x) = 2 - x^3$

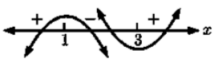
$\therefore\ f'(x) = -3x^2$ 

$\therefore\ f''(x) = -6x$ 

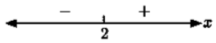
Now $f''(x) = 0$ when $x = 0$,
and $f'(0) = 0$
$\therefore$ there is a stationary inflection at $(0, 2)$.

c $f(x) = x^3 - 6x^2 + 9x + 1$

$\therefore\ f'(x) = 3x^2 - 12x + 9$
$= 3(x^2 - 4x + 3)$
$= 3(x - 3)(x - 1)$



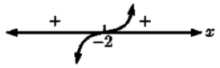
and $f''(x) = 6x - 12 = 6(x - 2)$



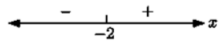
Now $f''(x) = 0$ when $x = 2$
and $f'(2) \neq 0$
$\therefore$ there is a non-stationary inflection at
$(2, f(2))$ which is $(2, 3)$.

d $f(x) = x^3 + 6x^2 + 12x + 5$

$\therefore\ f'(x) = 3x^2 + 12x + 12$
$= 3(x^2 + 4x + 4)$
$= 3(x + 2)^2$



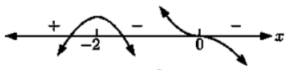
and $f''(x) = 6x + 12 = 6(x + 2)$



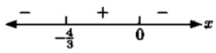
Now $f''(x) = 0$ when $x = -2$
and $f'(-2) = 0$
$\therefore$ there is a stationary inflection at
$(-2, f(-2))$ which is $(-2, -3)$.

e $f(x) = -3x^4 - 8x^3 + 2$

$\therefore\ f'(x) = -12x^3 - 24x^2$
$= -12x^2(x + 2)$



and $f''(x) = -36x^2 - 48x$
$= -12x(3x + 4)$



$\therefore$ there is a stationary inflection at $(0, 2)$,
and a non-stationary inflection at
$(-\frac{4}{3}, f(-\frac{4}{3}))$, which is $\left(-\frac{4}{3}, \frac{310}{27}\right)$.

f $f(x) = 3 - \dfrac{1}{\sqrt{x}} = 3 - x^{-\frac{1}{2}}$

$\therefore\ f'(x) = \frac{1}{2}x^{-\frac{3}{2}}$

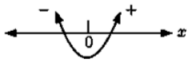
and $f''(x) = -\frac{3}{4}x^{-\frac{5}{2}} = \dfrac{-3}{4x^2\sqrt{x}}$

Now $f''(x) \neq 0$ for all x
$\therefore$ there are no points of inflection.

EXERCISE 37

a $f(x) = x^2$

$\therefore\ f'(x) = 2x$ which has sign diagram:



and $f''(x) = 2$

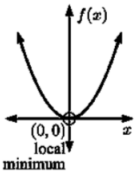
I There is a local minimum at $(0, 0)$.

II There are no points of inflection as $f''(x) \neq 0$.

III $f(x)$ is increasing when $x \geqslant 0$, and decreasing when $x \leqslant 0$.

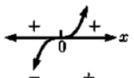
IV $f(x)$ is concave up for all x as $f''(x) > 0$ for all x.

V

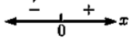


b $f(x) = x^3$

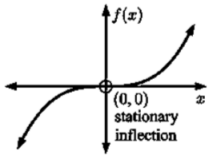
$\therefore\ f'(x) = 3x^2$ which has sign diagram:



and $f''(x) = 6x$ which has sign diagram:



V



I A stationary inflection at $(0, 0)$.

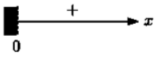
II A stationary inflection at $(0, 0)$.

III $f(x)$ is increasing for all real x.

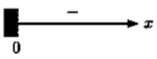
IV $f(x)$ is concave up when $x \geqslant 0$, and concave down when $x \leqslant 0$.

c $f(x) = \sqrt{x}$

$\therefore\ f'(x) = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$ which has sign diagram:



and $f''(x) = -\frac{1}{4}x^{-\frac{3}{2}} = \frac{-1}{4x\sqrt{x}}$ which has sign diagram:



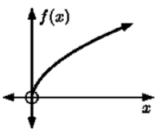
I There are no stationary points as $f'(x) \neq 0$.

II There are no points of inflection as $f''(x) \neq 0$.

III $f(x)$ is increasing for all $x \geqslant 0$, never decreasing.

IV $f(x)$ is concave down for all $x \geqslant 0$ as $f''(x) < 0$ for all $x > 0$, never concave up.

V

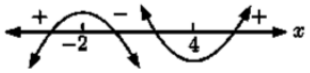


d $f(x) = x^3 - 3x^2 - 24x + 1$

$\therefore\ f'(x) = 3x^2 - 6x - 24$

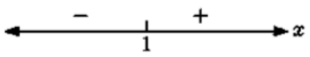
$= 3(x^2 - 2x - 8)$

$= 3(x - 4)(x + 2)$ which has sign diagram:



and $f''(x) = 6x - 6$

$= 6(x - 1)$ which has sign diagram:



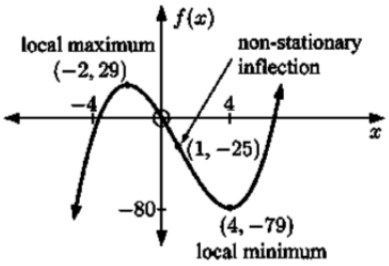
I $f(-2) = 29$, $f(4) = -79$, so there is a local maximum at $(-2, 29)$, and a local minimum at $(4, -79)$.

II $f(1) = -25$, so there is a non-stationary inflection at $(1, -25)$.

III $f(x)$ is increasing for $x \leqslant -2$ and $x \geqslant 4$, and decreasing for $-2 \leqslant x \leqslant 4$.

IV $f(x)$ is concave down for $x \leqslant 1$, and concave up for $x \geqslant 1$.

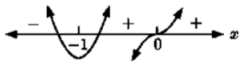
V



e $f(x) = 3x^4 + 4x^3 - 2$

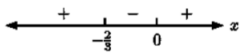
$\therefore \quad f'(x) = 12x^3 + 12x^2$

$= 12x^2(x+1)$ which has sign diagram:



and $f''(x) = 36x^2 + 24x$

$= 12x(3x+2)$ which has sign diagram:



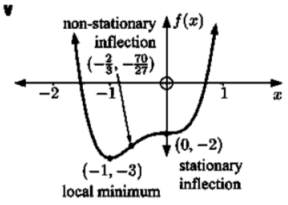
i There is a local minimum at $(-1, f(-1))$ which is $(-1, -3)$, and a stationary inflection at $(0, -2)$.

ii There is a non-stationary inflection at $(-\frac{2}{3}, f(-\frac{2}{3}))$ which is $(-\frac{2}{3}, -\frac{70}{27})$, and a stationary inflection at $(0, -2)$.

iii $f(x)$ is increasing for $x \geqslant -1$, and decreasing for $x \leqslant -1$.

iv $f(x)$ is concave down for $-\frac{2}{3} \leqslant x \leqslant 0$, and concave up for $x \leqslant -\frac{2}{3}$ and $x \geqslant 0$.

v

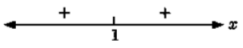


f $f(x) = (x-1)^4$

$\therefore \quad f'(x) = 4(x-1)^3$ which has sign diagram:



and $f''(x) = 12(x-1)^2$ which has sign diagram:



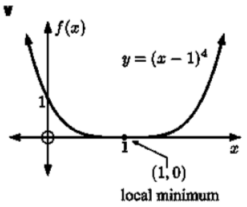
i There is a local minimum at $(1, 0)$.

ii Since there is no sign change in $f''(x)$ at $x = 1$, there are no points of inflection.

iii $f(x)$ is increasing for $x \geqslant 1$, and decreasing for $x \leqslant 1$.

iv $f(x)$ is concave up for all x.

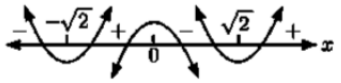
v



g $f(x) = x^4 - 4x^2 + 3$

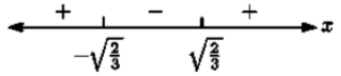
$f'(x) = 4x^3 - 8x = 4x(x^2 - 2)$

$= 4x(x+\sqrt{2})(x-\sqrt{2})$ which has sign diagram:



$f''(x) = 12x^2 - 8 = 4(3x^2 - 2)$

$= 4(\sqrt{3}x + \sqrt{2})(\sqrt{3}x - \sqrt{2})$ which has sign diagram:



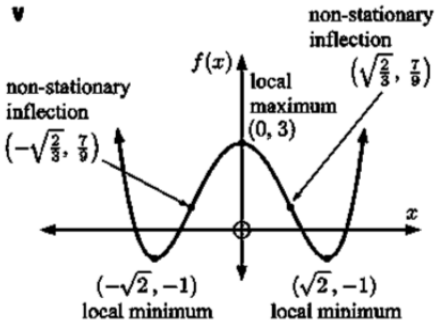
i There is a local maximum at $(0, 3)$, and $f(-\sqrt{2}) = f(\sqrt{2}) = -1$, so there are local minima at $(\sqrt{2}, -1)$ and $(-\sqrt{2}, -1)$.

ii $f\left(\sqrt{\frac{2}{3}}\right) = f\left(-\sqrt{\frac{2}{3}}\right) = \frac{7}{9}$, so there are non-stationary inflections at $(\sqrt{\frac{2}{3}}, \frac{7}{9})$ and $(-\sqrt{\frac{2}{3}}, \frac{7}{9})$.

iii $f(x)$ is increasing for $-\sqrt{2} \leqslant x \leqslant 0$ and $x \geqslant \sqrt{2}$, and decreasing for $x \leqslant -\sqrt{2}$ and $0 \leqslant x \leqslant \sqrt{2}$.

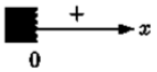
iv $f(x)$ is concave down for $-\sqrt{\frac{2}{3}} \leqslant x \leqslant \sqrt{\frac{2}{3}}$, and concave up for $x \leqslant -\sqrt{\frac{2}{3}}$ and $x \geqslant \sqrt{\frac{2}{3}}$.

v

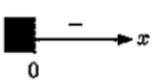


h $f(x) = 3 - \frac{4}{\sqrt{x}} = 3 - 4x^{-\frac{1}{2}}, \quad x > 0$

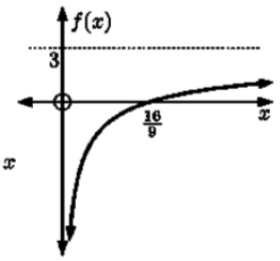
$\therefore \quad f'(x) = 2x^{-\frac{3}{2}} = \frac{2}{x\sqrt{x}}$ with sign diagram:



and $f''(x) = -3x^{-\frac{5}{2}} = -\frac{3}{x^2\sqrt{x}}$ with sign diagram:



v



i There are no stationary points as $f'(x) \neq 0$.

ii There are no points of inflection as $f''(x) \neq 0$.

iii $f(x)$ is increasing for all $x > 0$ and never decreasing.

iv $f(x)$ is concave down for all $x > 0$ and never concave up.

EXERCISE 38

EXERCISE 39

a The x-intercepts occur when $y = 0$

For $f(x) = e^x - 3$, $e^x - 3 = 0$

$\therefore \quad e^x = 3$

$\therefore \quad x = \ln 3$

and for $g(x) = 3 - \frac{5}{e^x}$, $3 - \frac{5}{e^x} = 0$

$\therefore \quad \frac{3e^x - 5}{e^x} = 0$

$\therefore \quad 3e^x - 5 = 0$

$\therefore \quad e^x = \frac{5}{3}$

$\therefore \quad x = \ln(\frac{5}{3})$

$\therefore$ $f(x)$ has x-intercept $\ln 3$

and $g(x)$ has x-intercept $\ln(\frac{5}{3})$.

The y-intercepts occur when $x = 0$

Now $f(0) = e^0 - 3 = -2$ and $g(0) = 3 - \frac{5}{e^0} = 3 - 5 = -2$

$\therefore$ both $f(x)$ and $g(x)$ have y-intercept -2.

b As $x \to \infty$, $f(x) \to \infty$ As $x \to \infty$, $g(x) \to 3^-$

$x \to -\infty$, $f(x) \to -3^+$ $x \to -\infty$, $g(x) \to -\infty$

c $f(x)$ and $g(x)$ meet when

$e^x - 3 = 3 - 5e^{-x}$

$\therefore \quad e^{2x} - 3e^x = 3e^x - 5 \quad \{\times e^x\}$

$\therefore \quad e^{2x} - 6e^x + 5 = 0$

$\therefore \quad (e^x - 5)(e^x - 1) = 0$

$\therefore \quad e^x = 5$ or 1

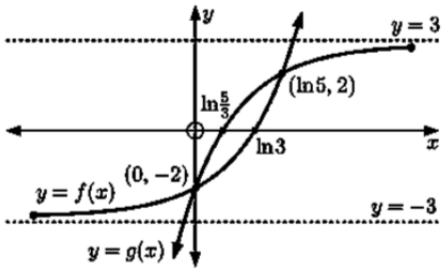
$\therefore \quad x = \ln 5$ or 0

Now $f(\ln 5) = e^{\ln 5} - 3 = 5 - 3 = 2$

and $f(0) = -2$

$\therefore$ $f(x)$ and $g(x)$ meet at $(\ln 5, 2)$ and $(0, -2)$.

d



EXERCISE 40

a Consider $y = e^x - 3e^{-x}$

It cuts the x-axis at P when $y = 0$

$\therefore \quad e^x - 3e^{-x} = 0$

$\therefore \quad e^{2x} - 3 = 0 \quad \{\times e^x\}$

$\therefore \quad e^{2x} = 3$

$\therefore \quad 2x = \ln 3$

$\therefore \quad x = \frac{1}{2}\ln 3$

It cuts the y-axis at Q when $x = 0$

$\therefore \quad y = e^0 - 3e^0$

$= 1 - 3$

$= -2$

$\therefore$ P is $(\frac{1}{2}\ln 3, 0)$ and Q is $(0, -2)$.

b $\dfrac{dy}{dx} = e^x + 3e^{-x}$

$= e^x + \dfrac{3}{e^x}$

Since $e^x > 0$ for all x,

$\dfrac{dy}{dx} > 0$ for all x

$\therefore$ the function is increasing for all x

c $\dfrac{dy}{dx} = e^x + 3e^{-x}$

$\therefore \quad \dfrac{d^2y}{dx^2} = e^x - 3e^{-x}$

$= y$

Above the x-axis $y > 0 \quad \therefore \quad \dfrac{d^2y}{dx^2} > 0$

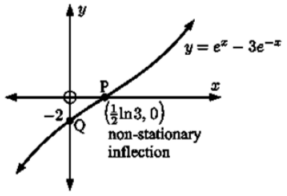
$\therefore$ the function is concave up

Below the x-axis $y < 0 \quad \therefore \quad \dfrac{d^2y}{dx^2} < 0$

$\therefore$ the function is concave down

$\therefore$ a non-stationary inflection occurs when $y = 0$

d



EXERCISE 41

$f(x) = \ln(2x - 1) - 3$

a $f(x) = 0$ when $\ln(2x - 1) = 3$

$\therefore \quad 2x - 1 = e^3$

$\therefore \quad 2x = e^3 + 1$

$\therefore \quad x = \dfrac{e^3 + 1}{2} \approx 10.5 \quad \therefore$ the x-intercept is $\dfrac{e^3 + 1}{2}$

b $f(0)$ cannot be found as $\ln(-1)$ is not defined. $\therefore$ there is no y-intercept.

c $f'(x) = \dfrac{2}{2x - 1} \quad \therefore \quad f'(1) = \dfrac{2}{2 - 1} = 2 \quad \therefore$ gradient of tangent $= 2$

d $\ln(2x - 1)$ has meaning provided $2x - 1 > 0 \quad \therefore \quad 2x > 1$ and so $x > \frac{1}{2}$

$\therefore$ the domain of f is $\{x \mid x > \frac{1}{2}\}$

e $f'(x) = 2(2x - 1)^{-1}$

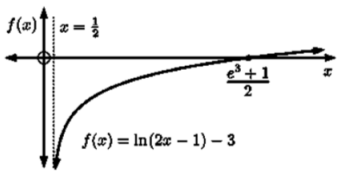
$\therefore \quad f''(x) = -2(2x - 1)^{-2}(2)$

$= \dfrac{-4}{(2x - 1)^2}, \quad x > \frac{1}{2}$

$\therefore$ provided $x > \frac{1}{2}, \quad f''(x) < 0$

$\therefore$ $f(x)$ is concave down for all x in the domain of f.

f



EXERCISE 42

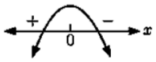
EXERCISE 43

a $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$

$\therefore \quad f'(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}(-x)$

$= \frac{-x}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$

$\therefore \quad f'(x) = 0$ when $x = 0$

$f'(x)$ has sign diagram: 

Now $f(0) = \frac{1}{\sqrt{2\pi}}$

so there is a local maximum at $\left(0, \frac{1}{\sqrt{2\pi}}\right)$.

The function is increasing for $x \leqslant 0$

and decreasing for $x \geqslant 0$

b $f'(x) = \frac{-x}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} = \frac{1}{\sqrt{2\pi}} \left(-xe^{-\frac{1}{2}x^2}\right)$

$\therefore \quad f''(x) = \frac{1}{\sqrt{2\pi}} \left((-1)\, e^{-\frac{1}{2}x^2} + (-x)\, e^{-\frac{1}{2}x^2}(-x)\right)$ {product rule}

$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \left(x^2 - 1\right)$

$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} (x+1)(x-1)$ which has sign diagram: 

Now $f(1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}} = \frac{1}{\sqrt{2e\pi}}$ and $f(-1) = \frac{1}{\sqrt{2e\pi}}$

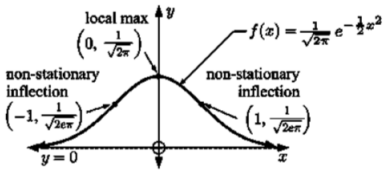
$\therefore$ there are non-stationary points of inflection at $\left(1, \frac{1}{\sqrt{2e\pi}}\right)$ and $\left(-1, \frac{1}{\sqrt{2e\pi}}\right)$.

c As $x \to \infty$, $e^{-\frac{1}{2}x^2} \to 0^+$

$\therefore \quad f(x) \to 0^+$

As $x \to -\infty$, $e^{-\frac{1}{2}x^2} \to 0^+$

$\therefore \quad f(x) \to 0^+$

d 

EXERCISE 44

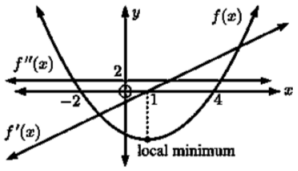
a $f(x)$ is quadratic, so $f'(x)$ will be linear and $f''(x)$ will be constant.

$f(x)$ is decreasing for $x \leqslant 1$ and increasing for $x \geqslant 1$

$\therefore \quad f'(x) \leqslant 0$ for $x \leqslant 1$ and $f'(x) \geqslant 0$ for $x \geqslant 1$

$\therefore \quad f'(x)$ is an increasing linear function which cuts the x-axis at 1.

As $f'(x)$ is increasing, $f''(x) > 0$.



b $f(x)$ is cubic, so $f'(x)$ will be quadratic and $f''(x)$ will be linear.

$f(x)$ has turning points at $x \approx \pm 1$

$\therefore \quad f'(x)$ cuts the x-axis at these points.

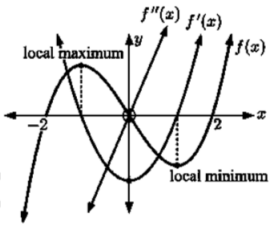
$f(x)$ has a non-stationary inflection point at $x = 0$

$\therefore \quad f'(x)$ has a turning point at $x = 0$, and $f''(0) = 0$.

$f(x)$ is concave down for $x \leqslant 0$ and concave up for $x \geqslant 0$

$\therefore \quad f'(x)$ is decreasing for $x \leqslant 0$ and increasing for $x \geqslant 0$

and $f''(x) \leqslant 0$ for $x \leqslant 0$ and $\geqslant 0$ for $x \geqslant 0$.



c $f(x)$ is cubic, so $f'(x)$ will be quadratic and $f''(x)$ will be linear.

$f(x)$ has turning points at $x \approx 1$ and $x = 3$

$\therefore \quad f'(x)$ cuts the x-axis at these points.

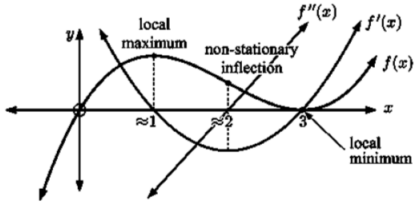
$f(x)$ has a non-stationary inflection point at $x \approx 2$

$\therefore \quad f'(x)$ has a turning point at $x \approx 2$, and $f''(2) = 0$

$f(x)$ is concave down for $x \leqslant 2$ and concave up for $x \geqslant 2$

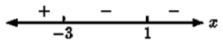
$\therefore \quad f'(x)$ is decreasing for $x \leqslant 2$ and increasing for $x \geqslant 2$

and $f''(x) \leqslant 0$ for $x \leqslant 2$ and $\geqslant 0$ for $x \geqslant 2$.



EXERCISE 45

a $f'(x)$ has sign diagram:



$\therefore$ $f(x)$ is increasing for $x \leqslant -3$, and decreasing for $x \geqslant -3$

$\therefore$ $f(x)$ has a local maximum at $x = -3$

$f'(x)$ has a turning point at $x \approx -1.7$.

At this point, $f''(x) = 0$, but $f'(x) \neq 0$

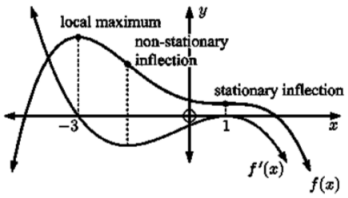
$\therefore$ $f(x)$ has a non-stationary inflection point here.

$f'(x)$ has another turning point at $x = 1$.

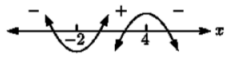
At this point, $f''(x) = 0$ and $f'(x) = 0$

$\therefore$ $f(x)$ has a stationary inflection point at $x = 1$.

A possible graph of $f(x)$ is shown alongside:



b $f'(x)$ has sign diagram:



$\therefore$ $f(x)$ has a local minimum at $x = -2$ and a local maximum at $x = 4$

$f'(x)$ has a turning point at $x \approx 1$.

At this point, $f''(x) = 0$, but $f'(x) \neq 0$

$\therefore$ $f(x)$ has a non-stationary inflection point at $x \approx 1$.

A possible graph of $f(x)$ is shown alongside:

